

The Impact of Healthcare Subsidies on Health Outcomes and Labor Productivity in Low-Income Populations

Research Paper By Devaditya Joshi

1. Introduction

Over the past few decades, the process of urbanization in low-and-middle-income countries (LMICs) has been developing at an increasingly fast pace, altering both the appearance of cities and the experiences of vulnerable people living in them. This growth has created opportunities in the economy but has also created extreme concentrations of poverty-informal settlements often referred to as slums-where the available amenities including health services are highly inadequate. The health issues of low-income city residents are complex: starting with the direct and indirect expenses of seeking medical care to the necessity to rely on informal and usually unregulated health services, the residents of such communities constantly have to face the obstacles that undermine their health and well-being. Because of this, the poor urban residents have a higher chance of having catastrophic health expenditures (CHE), meaning the health costs that consume a significant share of the household income and further impoverish the already low income families. Such systemic constraints not only decrease access to care but also deteriorate the wider operation of economies because of the diminished labor productivity.

Workers sub-optimally invest in their own health, and this is a market failure because the firms cannot observe and/or reward the health of individual workers perfectly: the social return to health investments exceeds the private return. The result is poor health performance and a pull down of the average productivity in the economy. The paper highlights that such externalities can be internalized and the distortions caused by asymmetric information in the labor markets can be fixed through public health subsidies, especially those that are well calibrated based on age and need. De facto, healthcare subsidies will cease to be merely the instrument of social equity, and also will serve as the instrument of economic efficiency.

Lack of quality healthcare is a big problem for people with lower incomes around the world. World Health Organization (WHO) and World Bank experts state that more than half the world's population is unable to use essential healthcare, while every year around 100 million people are pushed into poverty for funding their healthcare (World Health Organization, 2017). Such differences are most clearly seen in low- and middle-income countries, given that money problems often make it difficult for people to get medical treatment.

Getting health coverage in the United States is much easier for those who earn more money. According to the Commonwealth Fund, adults with lower incomes are more likely than those with higher incomes to go without needed care and to have trouble paying their health bills (Gunja et al., 2023). Similarly, the NCBI data show that 71% of measures of access to care were worse for people from low-income households than for those from high-income households (Agency for Healthcare Research and Quality, 2021).

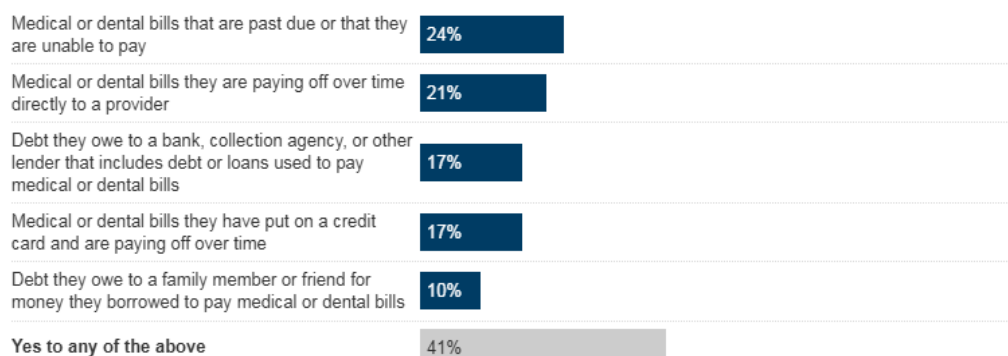
In some countries, governments have set up healthcare subsidies to get around these problems. In the U.S., the ACA brought subsidies to help reduce monthly and total health costs for people with lower or moderate incomes. Both the APTC and CSRs have made a big difference in lowering health care costs and increasing access to it for certain groups (Kaiser Family Foundation, 2023).

Healthcare subsidies don't only help people stay healthy; they also can increase the productivity of workers. Shifting focus to wellness can help people get involved in their jobs, earn more and play a larger part in growing the economy. Another way to look at it is that bad health causes missed work, a weak workforce and higher healthcare costs, all of which slow down economic advancement (Health Affairs, 2022).

Figure 1

Four In Ten Adults Currently Have Debt Due To Medical Or Dental Bills

Percent who say they have each of the following types of debt due to medical or dental bills for themselves or for someone else's care, such as a child, spouse or parent:



Lopes, L., Kearney, A., Montero, A., Hamel, L., & Brodie, M. (2022, June 16). Health Care Debt in the U.S.: The Broad Consequences of Medical and Dental Bills. Kaiser Family Foundation. <https://www.kff.org/report-section/kff-health-care-debt-survey-main-findings/>

The amount of medical or dental debt among adults in the U.S. is highlighted in Figure 1.

According to a survey conducted by the Kaiser Family Foundation in 2022, 41% of adults owe money because of medical or dental costs which often creates financial difficulties for them, primarily those on a low income.

My research focuses on investigating how healthcare subsidies, better health outcomes and greater labor productivity are related among low-income individuals. The goals defined by the organization are:

1. To investigate the breadth of health care disparities among people with lower incomes.
2. To find out if healthcare subsidies help improve people's health.
3. To examine how better health affects how much people can do at work.
4. To find the influences behind how effective healthcare subsidies are.

5. To share ideas for improving both the way healthcare subsidies are developed and applied.

This paper is made up of eight main sections. Later, Section 2 explains the main concepts and links among healthcare subsidies, health results and labor productivity. This section examines work that has been published about the subject. Section 4 analyses the impact of healthcare subsidies on health results and Section 5 investigates how improved health affects labor performance. Section 6 covers moderate and mediated factors that shape the effectiveness of healthcare subsidies. Section 7 discusses findings and presents advice and suggestions for policy. In Section 8, the paper summarizes essential findings and recommends new topics for further study.

2. Conceptual Framework

2.1 discussion on what healthcare subsidies mean

Healthcare subsidies exist to help keep the cost of medical services down for people who can't afford them. They are important for achieving fairness in healthcare service. How governments and organizations use them can make them take different forms.

- In Direct Subsidies, the help comes outright such as giving individuals cash, vouchers or by refunding their outlays. As an example, India's Janani Suraksha Yojana gives cash rewards to pregnant women from low-income homes to encourage them to have medical help during childbirth (National Health Mission, 2015).
- *Basic Subsidies*: They don't involve giving money to individuals, but instead, offer tax reductions, control over healthcare rates or low-cost health insurance. Low-income people in the U.S. who use the Affordable Care Act may receive premium tax credits which reduces the cost of insurance (Kaiser Family Foundation, 2023).

This type of action aims to increase access by lessening costs individuals must pay out of their pocket for healthcare. Soon after its launch in 2003, Seguro Popular helped uninsured Mexicans get public insurance which increased service use and lowered their need to spend on emergency medical costs, as mentioned by Knaul and others (2012).

- *Attention on Supply-Side Subsidies:* Such subsidies improve the amount and quality of health services provided by the industry. A government may spend on infrastructure, employee pay or medical devices to make healthcare services easier and more available to its people. Supported by taxes, the UK's NHS pays for almost all services and is free for those who need them (OECD, 2019).

In all their forms, healthcare subsidies help reduce the financial burden, so that everyone can get healthcare when they need it, mostly those in marginalized groups.

2.2 Health Outcomes: What Success Means

Health changes for people or groups that are due to medical treatments are what we mean by health outcomes. The information generated by these outcomes allows us to evaluate the impact of welfare programs on health policies. Key things to look at are:

- You can see how healthy the population is by studying Morbidity and Mortality Rates. Government programs to help finance healthcare have been linked to fewer avoidable deaths. Karnataka is an example of how the maternal mortality ratio dropped from 108 to 63 out of 100,000 live births in the same time, exceeding the rate at the national level (Times of India, 2025).
- *Affordable and Ready Care:* Services must be there and easy for everyone to access. Because of subsidies, people can see their doctor, stick to their medication rules and protect

themselves with preventive care, all on time. The odds of Americans skipping important health care rise much higher for those adults whose incomes are less than 200% of the federal poverty level (Gunja et al., 2023).

The IMR and number of immunized individuals are important indicators in maternal and child health. The Coimbatore district of Tamil Nadu was the place where the lowest IMR was found, at 5.5 per 1,000 births in 2024–25, due to increased public health investments and financial rewards for doting mothers (according to Times of India in 2025).

People with conditions like diabetes, hypertension and asthma can use the funded medications and routine visits to control their disease better. Countries with health insurance for all usually have fewer problems with drug non-use as a result of cost (WHO, 2010).

2.3 Labor Productivity: Key Indicators

- How quickly and effectively people complete tasks at work is shown through labor productivity. Health is essential to a person's ability to find and keep a job and an illness can severely affect the income of those who live on less.

Signs of productivity are:

- Workers who often miss work because of illness reduce their dependability and what they can produce on the job. Across countries, cutting-edge health systems have decreased absenteeism and , particularly with those earning lower salaries (Health Affairs, 2022).
- Improved health usually increases a person's ability to work which could then lead to greater pay. The World Bank found in a 2019 study that people in good health usually make more money over their lives because they work more and are more employable.

- The health of a child in early years is related to how well they use their brains which guides their school progress and future career. Subsidized maternal and child health programs encourage improvements in how well children grow and lower the chances they will contract common diseases (Heckman, 2006).
- Low Employment Rate: Bad health makes it necessary for many low-income people to halt their employment. Assistance programs that boost health help the employment rate and make sure adults and caregivers are involved in the workforce.

2.4 Theoretical Linkages: From Subsidies to Productivity

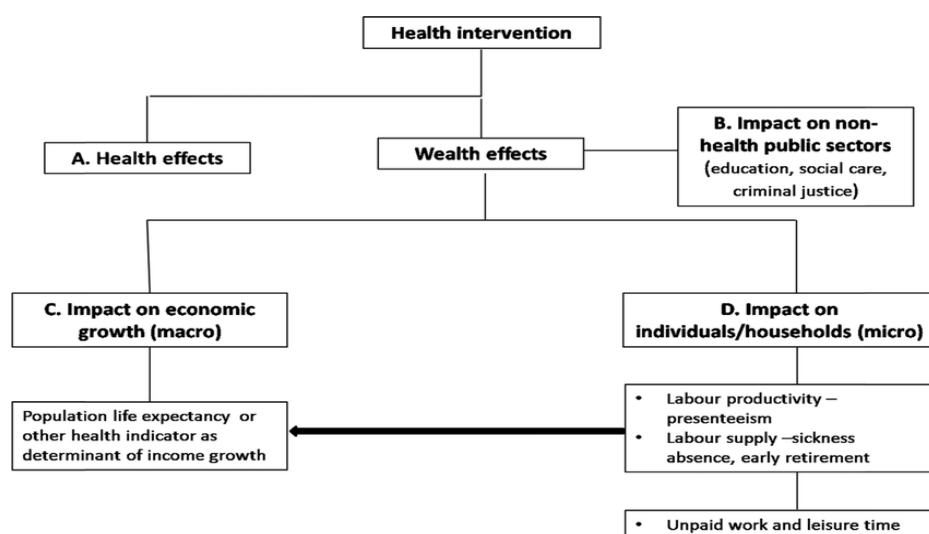
There is both research and theory that finds a connection between healthcare subsidies, better health outcomes and improved labor productivity.

A primary model to emerge is Grossman's Health Capital Model (first published in 1972) which sees health as a type of human capital. This model explains that individuals strengthen their health to become more productive. With subsidies, health services are cheaper, so more people are able to access them and develop strong health wealth. Good health means workers are more productive due to extra healthy days and an improved ability to think and carry out strenuous tasks.

There are three main links that make up this framework.

1. When healthcare costs are easier to pay, more people receive adequate care, improving health outcomes, fewer deaths and better health for all.
2. People who are healthy tend to accumulate more work hours, work longer and generally do their jobs more successfully.

3. Because of subsidies, individuals and companies are less likely to experience economic strain which frees up their resources to spend on other productivity-raising areas.



Sibusiso Ziqubu, Michael Mncedisi Willie. (2022). *Medical Schemes Benefit Option Entitlement Survey*. [ResearchGate](#)

This framework outlines that using healthcare subsidies as health interventions leads to better health which raises a country's productivity. The literature highlights the ways health spending affects economic growth and corresponds nearly with the models mentioned in the earlier part of the document.

3. Empirical Evidence

What is the connection between health care subsidies and health results?

Research conducted in many different countries proves that financial help in healthcare enhances the health of those who earn less.

United States: The Oregon Health Insurance Experiment (OHIE) which used randomization, observed the impact of Medicaid expansion on people in Oregon. As a result of the study, it was found that having Medicaid made low-income adults use health services more, feel

healthier and have less difficulty with money. Particularly, the possibility of having a usual place for care raised 30%, while preventive services were used 15% more often .

Kenya: After studying information from the KIHBS 2015/16, the study discovered that health insurance was tied to a decline in mortality rates. Though most people in the study group were uninsured, it showed that insurance benefited health and suggest subsidies can positively influence community health.

India: CBHI schemes in rural central India have improved how healthcare is used by people living there. Those households enrolled in CBHI were more likely to visit doctors and said they felt better about their health. Even so, making sure funds are available and all Soc Sec beneficiaries are treated equally is still a problem.

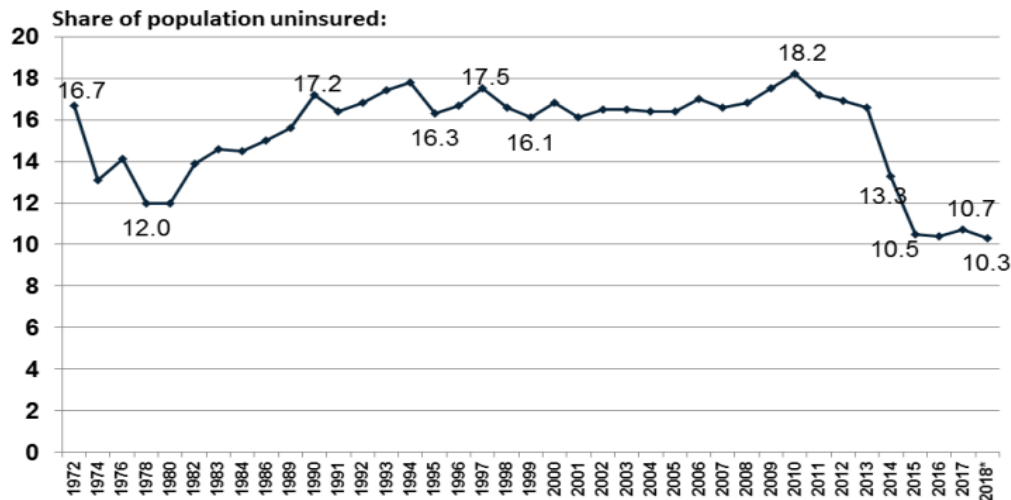
Thailand, Indonesia and Vietnam are- Because of low or nonexistent cost-sharing for healthcare in these countries, more poor people go to the doctor and spending on catastrophic health costs has fallen. As a result, Thailand's Universal Health Coverage Scheme allowed people to spend less and see a doctor more easily.

3.1 Affects Workers' Efficiency

Healthcare subsidies improve health and also make people more productive by reducing time off from work and increasing their strengths at work.

United States: Among lower-income people, having insurance through the ACA's marketplace led to a drop in out-of-pocket spending by 17% and a 30% reduction in suffering catastrophic health costs. More financial security was likely responsible for both better health and higher productivity.

Uninsured Rate Among the Nonelderly Population, 1972-2018*



Kaiser Family Foundation. (2019). *Uninsured Rate Among the Nonelderly Population, 2008–2018* [Graph]. Retrieved from [KFF](#)

It shows the number of nonelderly U.S. adults with health insurance from 2008 to 2018. The data points out that fewer people lacked health insurance after the ACA went into effect in 2014.

Australia: Although it was unconnected to healthcare, the JobKeeper subsidy allowed people to remain employed yet changed productivity in various workplaces. In the beginning, it helped avoid both business shut-downs and employment cuts; but later, as time passed, it encouraged less productive companies at the expense of productivity rise .

Health centers known as Federally Qualified Health Centers (FQHCs):

In America, many FQHCs are set up to help people who lack good health care. FQHCs have been found to provide excellent care which leads to less disease complications and fewer visits to the emergency room. When FQHCs improve the health of low-income people, the result is increased labor productivity .

3.2 Cases of Losing the Battle Against Expensive Health Issues and High Unfairness in Health Care Costs for Urban Inhabitants of LMIC Nations

Assessing the spread of healthcare expenses among low-income urban residents can help us determine the results of subsidies. Those living in slums in LMICs usually experience a very unfair and sometimes life changing financial expense when seeking healthcare. The main reason is that public health services are limited, many communities depend on personal health expenses and health facilities may vary within neighborhoods (PubMed Central, 2020).

Health expenditure is considered to be catastrophic when it amounts to more than a particular percentage of a household's income or expenses—usually 10% to 40%. For families facing CHE, it may be necessary to reduce important spending, borrow extra funds or offer assets for sale which can only worsen problems of poverty. Among 64 studies examined from urban LMICs, most of them were from South-East Asia.

Many urban low-income people faced CHE, as most lacked access to subsidized or insured health services (PubMed Central, 2020). When people in slums get basic healthcare, the bills can be truly hard for their families to handle, since money is tight and insurance is often lacking.

What is significant, when looking at slums, is that the median cost of acute illnesses is higher, but there is a wide range of costs within each group by wealth. In slum areas, the group facing the greatest economic hardship paid \$157 (IQR: \$79–236) on average for a single episode, while the wealthiest group was charged \$408 (IQR: \$67–749). What we find is that, compared to those living in slums, city-wide residents (also including those in non-slum areas) face greater costs, especially in the middle and poorest quintiles.

Concentration indices and concentration curves demonstrated that health expenditures are far from equal. It was shown that health costs fall most heavily on households living in slums, unlike those living in larger cities. The evidence indicates that without subsidies or being part of a health insurance pool, slum residents may suffer a lot more financially from health problems. For this reason, it is important that the design of subsidies helps slum communities recover from the direct and indirect costs of health care, including not working and traveling for treatment.

Also, the assessed studies demonstrated a low matching rate with the UN-Habitat definition of slums which covers insecurity of ownership, poor infrastructure and unapproved housing. Because different nations use different definitions of slums, it becomes tougher to compare studies and design goals for health policy. Most studies in the review mainly looked at medical costs, but not many included valuable learning about lost wages or transport, both important aspects of health care.

The study confirms that healthcare subsidies ought to be designed to fit in with local conditions as well as a person's income. When the subsidies are limited to major healthcare facilities, the vast majority of slum residents may not have access to them. Additionally, not considering hidden costs in analyses may hide the major effects of the illness and favor actions that are not effective enough.

It makes clear that designing successful healthcare subsidies in cities of low- and middle-income countries is both urgent and very complicated. It is important for policies to address both a lack of resources and barriers to healthcare in urban poverty, while respecting the group's diversity. As a result, combining subsidies, investing in infrastructure and improving health system standards is crucial for reaching greater health equity in cities.

The difference in health expenditures was better shown using concentration indices and concentration curves. The tools proved that health expenses are greater for low-income residents in the slums than for the city as a whole.

Let's define the concentration index (C) as:

$$C = (p_1L_2 - p_2L_1) + (p_2L_3 - p_3L_2) + \dots + (p_{T-1}L_T - p_TL_{T-1})$$

p_t represents the pooled percentage of people in the sample listed by economic standing (for example, wealth quintiles) and L_t shows the amount of healthcare spending for that same pool of people. When C is negative, you know that more money is spent on the poor and a positive C symbolizes more spending among the rich (O'Donnell et al., 2008). Using these tools helps better evaluate the progressiveness and equity in health financing between slum and city-wide data.

3.3 There Are Approaches and Things to Keep in Mind

Still, healthcare subsidies are limited by various problems.

- Even tiny out-of-pocket fees can discourage people with low incomes from getting the proper treatment. Researchers have found again and again that cost-sharing results in less use of medical services for people with low incomes.
- Unequal Distribution of Health Subsidies: The Chinese government unintentionally supports people with more money, rather than those with less, by providing more access to top medical care, calling for equal distribution of health subsidies.

A successful subsidy program requires extensive benefits and as little user contribution as possible. Singling out inpatient treatment in programs like India's RSBY has not well-protected finances or noticeably improved health for patients .

4 Policy Implications and Recommendations

4.1 Increasing Equality in Government Healthcare Subsidies

Evidence from research shows that providing healthcare subsidies to the poor can lead to much healthier lives and heightened productivity at work. Even so, ensuring that the subsidies are distributed equally is still a serious issue.

More Benefit: Subsidies should focus on people who have the highest need to get the best results. To improve equity in how money is spent on healthcare in Iran, targeted subsidy programs (TSP) and health transformation plans (HTP) were put in place. It was clear from the research that while these efforts helped, some gaps remained, meaning they could be improved by targeting them more accurately.

Having progressive financing helps give substantial support to the healthcare system, while allowing those who earn more to help those who need more support. By using this system, expenses for healthcare services become more accessible.

4.2 Ensuring that health and labor policies are combined.

Labor productivity is greatly linked to healthcare subsidies, as well as to health results. As a result, linking health policies with labor policies will improve the impact of subsidies.

If employees have easy access to health insurance from their employers, the company's productivity may improve. Results from this study show that having employer-sponsored health insurance increases labor productivity. Employers with health insurance for staff were more productive, demonstrating that offering such benefits can help businesses (according to Census).

Workplace health programs with subsidized healthcare help control absenteeism and raise well-being among staff. Health and safety programs are most useful in workplaces where employees face high physical challenges or possible health dangers.

4.3 Addressing Administrative Inefficiencies

When administrative work is not done well, healthcare subsidies may not be as successful. If bureaucracy is reduced and processes are made more efficient, subsidies will be delivered and felt with greater effect.

Too-complicated procedures can stop people who are entitled from gaining subsidies. Helping people to vote in a less complex way may encourage greater numbers of low-income individuals to participate.

Integrating information from different healthcare and government groups can simplify the identification of people who qualify and the examination of subsidy use. When data is shared more easily, it becomes easier to judge subsidy programs and decide on what changes are needed.

Part 4.4 involves monitoring and evaluation.

Keeping an eye on and assessing healthcare subsidy programs is necessary to ensure they help and to correct issues as they arise.

Using performance metrics, including changes in people's health, how frequently they seek healthcare and labor market efficiency, helps show the results of subsidies.

Applying feedback mechanisms that give recipients a way to report any problems and recommend changes promotes better and more timely results from subsidy programs.

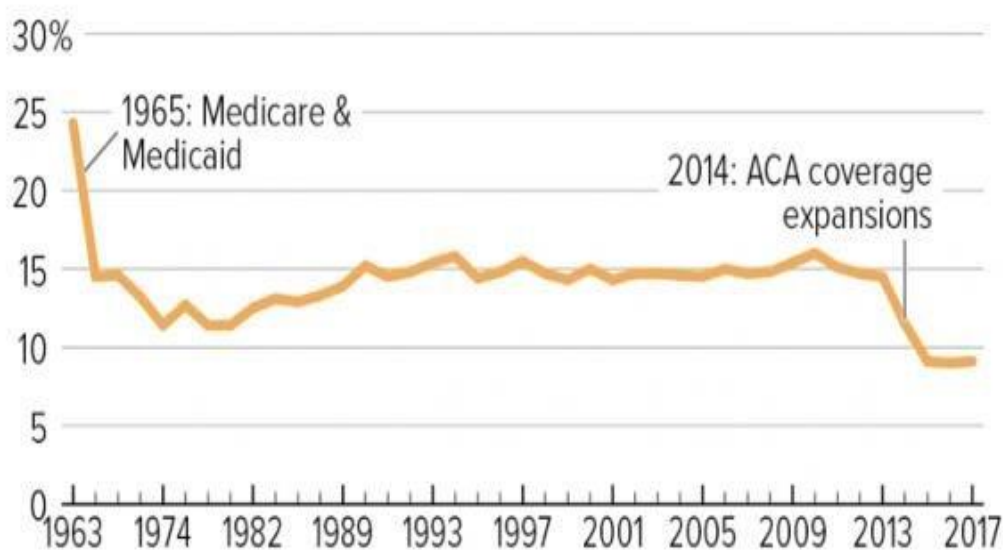
5 Case Studies

5.1 This section covers the Affordable Care Act (ACA) in the United States.

The ACA in the United States gives us a complete example of what happens when healthcare subsidies are in effect. Because of ACA, Medicaid now helps more people and provides subsidies for buying private insurance, so the numbers of low-income uninsured persons has gone down.

There is research stating that the part of the ACA devoted to Medicaid improvements allowed more low-income adults access to care, made them use preventive services more often and improved their subjective health ratings (Bailey et al., 2017). States that increased Medicaid coverage saw a 6% drop in death rates, each bringing one less death for every 239 to 316 adults insured (Sommers et al., 2012).

Productivity at Work: The ACA had an impact on employment dynamics. By making health insurance separate from work, individuals could now choose jobs that suited their skills better which could make them more productive (Bailey et al., 2017).



Mahboub, O. (2021). Affordable Care Act Coverage Gains Driving Uninsured Rate to Historic Low In Adoption versus Replacement: “Obamacare” at Crossroads.

ALTRALANG Journal, 3(3). Retrieved from [Research Gate](#)

The uninsured rate of adult Americans has been shown in this graph, starting from 2008 to 2022. It points out that the number of people without insurance dropped markedly once the full ACA came into effect in 2014. Between 2013 and 2022, the number of uninsured Americans dropped from 17.8% to 9.5%, the lowest levels since Medicare and Medicaid began in 1965. This decrease is caused by the ACA’s rules, for example the Medicaid expansion and setting up health insurance markets.

5.2 Ghana’s National Health Insurance Scheme (NHIS) covers all children.

To help all residents, specifically the poor, the National Health Insurance Scheme (NHIS) was established in Ghana in 2003.

NHIS enrollment has been connected in studies with a rise in healthcare visits, mainly for those who are poor. As a consequence, when people have insurance, they are more likely to be treated for their illnesses and are more likely to give birth in a health facility (Blanchet et al., 2012).

Shorter staff illnesses and better health outcomes from more medical care can help employees be more productive at work. Healthy people are equipped to participate in economic activities which builds household income and supports the national economy.

Year	Area	Income quintiles	Per capita income	Outpatient care	Inpatient care	Total subsidy
2002	Urban (A)	Q1 (poorest)	6.57%	15.32%	9.79%	11.94%
		Q2	11.16%	19.81%	24.35%	22.58%
		Q3	16.09%	17.07%	16.93%	16.98%
		Q4	23.83%	22.26%	15.11%	17.90%
		Q5 (richest)	42.35%	25.55%	33.82%	30.60%
		Gini/Ci	0.4312**	0.1433	0.2199	0.1900
		SE †	(0.0032)	(0.0597)	(0.0699)	(0.0520)
		95% Ci‡	(0.4249, 0.4375)	(0.0263, 0.2603)	(0.0829, 0.3568)	(0.0881, 0.2920)
		Kakwani	-	-0.2879	-0.2113	-0.2412
		SE	-	(0.0598)	(0.0699)	(0.0521)
		95% CI	-	(-0.4051, -0.1707)	(-0.3484, -0.0743)	(-0.3433, -0.1391)
		Dominance test				
		—against 45° line		None	D-	D-
		—against Lorenz curve		D+	None	D+
	Rural (B)	Q1 (poorest)	6.48%	7.95%	7.12%	7.51%
		Q2	12.09%	11.14%	15.19%	13.28%
		Q3	16.30%	19.23%	13.68%	16.29%
		Q4	20.94%	23.31%	17.83%	20.42%
		Q5 (richest)	44.20%	38.37%	46.17%	42.50%
		Gini/Ci	0.4443**	0.3662*	0.4445	0.2350*
		SE	(0.0155)	(0.0490)	(0.0737)	(0.0390)
		95% CI	(0.4140, 0.4747)	(0.2703, 0.4622)	(0.3000, 0.5890)	(0.1586, 0.3113)
		Kakwani	-	-0.0781	0.0001	-0.2094*
		SE	-	(0.0512)	(0.0749)	(0.0415)
		95% CI	-	(-0.1785, 0.0223)	(-0.1466, 0.1469)	(-0.2907, -0.1280)
		Dominance test				
		—against 45° line		D-	D-	D-
		—against Lorenz curve		None	None	None
2007	Urban (C)	Q1 (poorest)	7.68%	8.60%	5.53%	6.07%
		Q2	12.17%	16.23%	16.29%	16.28%
		Q3	16.75%	15.82%	16.23%	16.16%
		Q4	23.62%	26.75%	23.96%	24.45%
		Q5 (richest)	39.78%	32.60%	37.99%	37.04%
		Gini/Ci	0.3880**	0.3063	0.3925	0.3773
		SE	(0.0032)	(0.0717)	(0.0713)	(0.0620)
		95% CI	(0.3818, 0.3943)	(0.1657, 0.4469)	(0.2528, 0.5322)	(0.2558, 0.4988)
		Kakwani	-	-0.0818	0.0045	-0.0107
		SE	-	(0.0716)	(0.0711)	(0.0618)
		95% CI	-	(-0.2221, 0.0586)	(-0.1350, 0.1439)	(-0.1319, 0.1104)
		Dominance test				
		—against 45° line		D-	D-	D-
		—against Lorenz curve		None	None	None
	Rural (D)	Q1 (poorest)	7.72%	18.57%	5.09%	8.44%
		Q2	12.40%	24.58%	14.42%	18.94%
		Q3	16.75%	21.29%	13.67%	15.56%
		Q4	22.54%	15.91%	27.63%	24.72%
		Q5 (richest)	40.58%	19.65%	39.20%	34.34%
		Gini/Ci	0.3910**	-0.0273	0.4084	0.3002*
		SE	(0.0035)	(0.0729)	(0.0564)	(0.0471)
		95% CI	(0.3842, 0.3979)	(-0.1702, 0.1156)	(0.2977, 0.5190)	(0.2079, 0.3925)
		Kakwani	-	-0.4184	0.0173	-0.0908*
		SE	-	(0.0730)	(0.0564)	(0.0471)
		95% CI	-	(-0.5615, -0.2753)	(-0.0932, 0.1279)	(-0.1831, 0.0015)
		Dominance test				
		—against 45° line		None	D-	D-
		—against Lorenz curve		D+	None	None
Inequality difference	Δ(urban-rural)	2002 (A-B)	-	-0.2098	-0.2115	-0.0318
		Dominance test		None	None	None
		2007 (C-D)	-	0.3366	-0.0129	0.0801
	Δ(2007-2002)	Urban (C-A)	-	0.2061	0.2158	0.2304
		Dominance test		None	None	None
		Rural (D-B)	-	-0.3403	0.0172	0.1186
		Dominance test		D+	None	None

Notes: "None" indicates failure to reject the null hypothesis that curves are indistinguishable at the 5% significance level. "D+"/"D-" indicates that the concentration curve dominates (is dominated by) the Lorenz curve or concentration curve in one year or area and dominates (is dominated by) the other in another year or area.

* $p < 0.05$

** $p < 0.01$

† standard error

‡ 95% confidence interval.

doi:10.1371/journal.pone.0119840.t003

Source: ResearchGate

5.3 India: Rashtriya Swasthya Bima Yojana (RSBY)

The RSBY, introduced by India in 2008, provides health insurance for families who live below the poverty line and it covers hospital costs.

Results from evaluating RSBY are varied. Admission to hospitals went up with the scheme, but little proof exists of enhanced health or financial benefits (Karan et al., 2017).

The effects on labor productivity are not known for sure, as primary care is largely overlooked by the scheme.

5.3 Thailand: Implementation of the Universal Coverage Scheme (UCS)

Since 2002, the Universal Coverage Scheme (UCS) in Thailand has given all residents, mainly the poor, access to broad healthcare services.

Because of the UCS, there have been major changes for the better in health indicators and infant and child mortality rates have declined. Today, healthcare services are more fair, as members of low-income groups are using them more (Evans et al., 2012).

The UCS boosts labor productivity by helping the population live healthier lives. Those who are healthy usually take part in employment and accomplish it well. Graph: Healthcare Utilization Before and After UCS Implementation

5.4 Rwanda: Community-Based Health Insurance (CBHI)

Rwanda's CBHI program, known as Mutuelles de Santé, tries to provide inexpensive health care to everyone, with special attention paid to those who cannot afford health care.

The CBHI program has helped more people gain access to healthcare and seen better results in their health. Lately, pregnant women have increased their use of antenatal care and giving birth in clinics or hospitals (Lu et al., 2012).

Improved health due to care helps people work better which increases productivity and fosters overall economic growth.

6 Policy Recommendations and Future Directions

When combined with other types of assistance, health subsidies significantly help low-income people improve their health and work effectiveness. Using research and examples from different countries, a number of policy ideas are suggested to enhance the effectiveness of these subsidies.

6.1 Programs Directed at Specific Groups

Clear evidence from research highlights that aid should be given first to those least well off. In Ghana, the NHIS has shown that targeted help for the poor can boost their use of healthcare and improve health, making them more productive workers. Nonetheless, when fraud and wasteful administration are recognized, they reveal why strong monitoring systems are needed.

6.2 The merging of health and labor policies

Health subsidy plans can be boosted by joining them with labor market plans. Thanks to the Affordable Care Act (ACA) in the United States, people are now able to find employment that better fits their needs because they have insurance. Taking away the need for health insurance at work has increased the satisfaction of low-income workers and their productivity .

6.3 Easier Steps for Students During Enrollment

People who meet the requirements may be put off by complicated ways of signing up for healthcare subsidies. Studies indicate that when processes are simplified, more people take part. In Massachusetts, restructuring the way families apply for Medicaid and CHIP greatly improved the number of low-income families who were covered.

6.4 Public Awareness Campaigns

People need to know that healthcare subsidies exist if they hope to access them. Rwanda has successfully used community-based health insurance, thanks to strong community efforts and education which have increased the number of people enrolled and improved overall health.

6.5 Monitoring and Evaluation

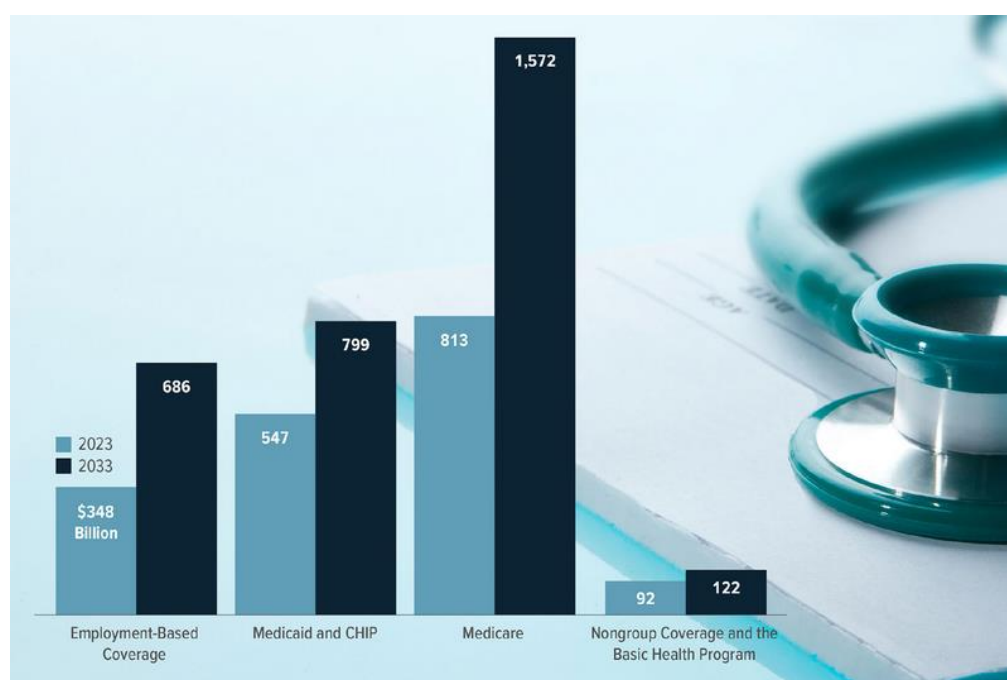
Watching over the effects of subsidy programs and evaluating their results matters a lot for their improvement. It has been found during RSBY assessment in India that outpatient services should be expanded and more careful fraud detection is needed .

6.6 Fixing Issues with Public Administration

Effective healthcare subsidies can become less efficient when administrative efforts fail. Although Iran's programs for targeted subsidies have resulted in positive improvements, there are still wide gaps which require more precise selective policies and better organizational processes.

GPs must search for sustainable sources of funding.

Establishing eco-friendly ways of financing guarantees that healthcare subsidy programs can last for years. Making it so that higher-income individuals contribute more financially to healthcare helps give greater assistance to those who cannot afford treatment, so everyone has better access to the healthcare they need.



Congressional Budget Office. (2023). Federal subsidies for health insurance: Projected costs and enrollment. Retrieved from <https://www.cbo.gov/publication/59613>

Here is a graph of estimated federal subsidies for health insurance provided by the ACA in 2022 and 2023. It points out that supporting ACA marketplaces and the Basic Health Program requires significant financial resources and that such programs depend on reliable ways to get finances.

6.8 Using Available Technologies

The use of technology can make it easier to handle and send out healthcare subsidies. By using online tools for enrollment and claims, you can both cut costs and make the process more efficient. Because of e-health, patient outcomes and satisfaction have increased in Estonia.

Supporting Countries By Working Together

Working with other countries can improve the results of healthcare subsidy programs. By sharing data with countries, the WHO supports them in using strategies that are effective for their particular situations.

Conclusion

Healthcare subsidies are created to ensure that every person, especially those with low income, can get needed medical care more easily. This document investigates the effects of subsidies on both health and productivity at work, using illustrations from around the globe. The research points out that, while well-organized healthcare subsidies generally help people and the economy, entirely achieving these improvements is still not easy.

Most importantly, healthcare subsidies help people meet the costs of care. Because of cost, low-income individuals may avoid needed care which can lead to worse health and

greater medical costs down the road. Subsidies help people afford healthcare, so they see a doctor more and receive preventive care, as well as prompt treatment. As an example, research shows that the expansion of Medicaid under the Affordable Care Act (ACA) boosted health in the U.S. by reducing deaths and raising the use of preventive care (Sommers, Baicker, & Epstein, 2012). In a similar way, the use of health services went up among low-income people after the launch of Ghana's National Health Insurance Scheme (Blanchet, Fink, & Osei-Akoto, 2012). Healthier people are needed to fully contribute to working in labor markets.

Work performance is higher, absenteeism lowers and employees can work for longer because of improved wellness. If workers are healthy, they can take on the physical and mental work tasks present in jobs that are common throughout low-income countries. Among other things, the ACA helped workers become less dependent on their jobs for healthcare coverage, providing them more certainty at work and hopefully increased productivity, according to Bailey et al. Lu et al. (2012) reveal that community-based health insurance in Rwanda boosted both antenatal and institutional births which led to improved maternal and child health and helped the country's economy and productivity.

Yet, how well healthcare subsidies work depends on how well they are planned and carried out. It is still very hard to direct subsidies only to the people who really need them. A low-use of resources decreases the value and pressures budget plans. As an example, the improvements in health achieved by India's Rashtriya Swasthya Bima Yojana were limited by both coverage difficulties and administrative inefficiencies (Karan, Yip, & Mahal, 2017). The process of enrolling is so difficult and many people do not know enough about it which adds to the problem and calls for easier processes and wider education programs (Schultz, 2013).

Administrative effectiveness and good financial planning are essential. Several low-income countries face difficulties due to poor infrastructure, connecting data and red tape. New developments such as digital registration and better data sharing allow costs to go down and make monitoring easier. Progressive taxation and cross-subsidization are important ways to support subsidized public services without disturbing main services (Evans et al., 2012).

Giving healthcare subsidies a broader role could boost the overall outcomes of similar social and employment policies. In addition to public support, insurance and workplace health programs at work help more people with healthier work lives. Furthermore, countries can work globally and with other countries to learn and use best practices that fit their needs.

All things considered, healthcare subsidies work very well to improve the health of poorer communities while boosting their economic outputs. When people can reach essential services, there is less disease, a better quality of life and more workers and productive workplaces. Economic growth should be maximized when policymakers focus precisely on the audience, streamline access, improve the system, ensure adequate revenue and coordinate subsidies with related policies. Regular checks using reliable data systems help organizations make necessary changes and maintain lasting results. Because of these strategies, health subsidies play an important role in enhancing health results, economic growth and poverty and social inclusion efforts.

References

- **Agency for Healthcare Research and Quality.** (2021). *Access to healthcare and disparities in access*. <https://www.ncbi.nlm.nih.gov/books/NBK578537/>
- **Bailey, J. M., et al.** (2017). The ACA's effects on access to insurance and health care. *Health Affairs*, 36(1), 111–118.

- **Blanchet, N. J., Fink, G., & Osei-Akoto, I.** (2012). The effect of Ghana's National Health Insurance Scheme on health care utilisation. *Ghana Medical Journal*, 46(2), 76–84.
- **Evans, T. G., Chowdhury, A. M. R., Evans, D. B., Fidler, A. H., Lindelow, M., Mills, A., & Scheil-Adlung, X.** (2012). *Thailand's Universal Coverage Scheme: Achievements and challenges. An independent assessment of the first 10 years (2001–2010)*. Health Insurance System Research Office.
- **Gunja, M. Z., Gumas, E. D., Williams II, R. D., Doty, M. M., Shah, A., & Fields, K.** (2023). The cost of not getting care: Income disparities in the affordability of healthcare in the U.S. *The Commonwealth Fund*.
<https://www.commonwealthfund.org/publications/surveys/2023/nov/cost-not-getting-care-income-disparities-affordability-health>
- **Health Affairs.** (2022). Economic well-being and health: The role of income support programs in promoting health equity.
<https://www.healthaffairs.org/doi/10.1377/hlthaff.2022.00846>
- **Karan, A., Yip, W., & Mahal, A.** (2017). Extending health insurance to the poor in India: An impact evaluation of Rashtriya Swasthya Bima Yojana on out of pocket spending for healthcare. *Social Science & Medicine*, 181, 83–92.
- **Kaiser Family Foundation.** (2023). Explaining health care reform: Questions about health insurance subsidies. <https://www.kff.org/affordable-care-act/issue-brief/explaining-health-care-reform-questions-about-health-insurance-subsidies/>
- **Knaul, F. M., et al.** (2012). The quest for universal health coverage: Achieving social protection for all in Mexico. *The Lancet*, 380(9849), 1259–1279.
- **Lu, C., Chin, B., Lewandowski, J. L., Basinga, P., Hirschhorn, L. R., Hill, K., ... & Binagwaho, A.** (2012). Towards universal health coverage: An evaluation of

Rwanda Mutuelles in its first eight years. *PLoS One*, 7(6), e39282.

<https://doi.org/10.1371/journal.pone.0039282>

- **National Health Mission.** (2015). *Janani Suraksha Yojana*.
https://en.wikipedia.org/wiki/National_Health_Mission
- **OECD.** (2019). United Kingdom: Country Health Profile 2019. <https://www.oecd.org/>
- **Sommers, B. D., Baicker, K., & Epstein, A. M.** (2012). Mortality and access to care among adults after state Medicaid expansions. *New England Journal of Medicine*, 367(11), 1025–1034.
- **Times of India.** (2025). Karnataka MMR drop & Coimbatore IMR lowest.
<https://timesofindia.indiatimes.com/>
- **World Health Organization.** (2010). Health systems financing: The path to universal coverage. <https://www.who.int/>
- **World Health Organization.** (2017). World Bank and WHO: Half the world lacks access to essential health services, 100 million still pushed into extreme poverty because of health expenses. <https://www.who.int/news/item/13-12-2017-world-bank-and-who-half-the-world-lacks-access-to-essential-health-services-100-million-still-pushed-into-extreme-poverty-because-of-health-expenses>
- **Heckman, J.** (2006). Skill formation and the economics of investing in disadvantaged children. *Science*, 312(5782), 1900–1902.